DAA ASSIGNMENT

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CH.SC.U4CSE24250

HUFFMAN CODING: -

CODE: -

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    char data;
    int freq;
    struct Node *left, *right;
};

struct Node* createNode(char data, int freq) {
    struct Node* node = (struct Node*)malloc(sizeof(struct Node));
    node->data = data;
    node->freq = freq;
    node->left = node->right = NULL;
    return node;
}

void swap(struct Node **a, struct Node **b) {
    struct Node *t = *a;
    *a = *b;
    *b = t;
}
```

```
void sort(struct Node *arr[], int n) {
    for(int i=0;i<n-1;i++)
        for(int j=0;j<n-i-1;j++)
            if(arr[j]->freq > arr[j+1]->freq)
                swap(&arr[j], &arr[j+1]);
}

void printCodes(struct Node* root, int code[], int top) {

    if(root->left) {
        code[top] = 0;
        printCodes(root->left, code, top + 1);
    }

    if(root->right) {
        code[top] = 1;
        printCodes(root->right, code, top + 1);
    }

    if(root->left == NULL && root->right == NULL) {
        printf("%c : ", root->data);
        for(int i=0;i<top;i++)
            printf("%d", code[i]);
        printf("\n");
    }
}

int main() {

    int n;
```

```
printf("Enter number of characters: ");
scanf("%d",&n);

char ch[n];
int freq[n];

struct Node* arr[n];

printf("Enter characters:\n");
for(int i=0;i<n;i++)
   scanf(" %c",&ch[i]);

printf("Enter frequencies:\n");
for(int i=0;i<n;i++) {
   scanf("%d",&freq[i]);
   arr[i] = createNode(ch[i], freq[i]);
}

int size = n;

while(size > 1) {

   sort(arr, size);

   struct Node *left = arr[0];
   struct Node *right = arr[1];

   struct Node *newNode = createNode('$', left->freq + right->freq);
   newNode->left = left;
   newNode->right = right;
```

```
        arr[1] = newNode;

        for(int i=0;i<size-1;i++)
            arr[i] = arr[i+1];

        size--;
    }

    struct Node *root = arr[0];

    int code[20];

    printf("\nHuffman Codes:\n");
    printCodes(root, code, 0);

    return 0;
}
```

OUTPUT: -

```
bharg@HP:~$ nano huffman.c
bharg@HP:~$ gcc huffman.c -o huffman
bharg@HP:~$ ./huffman
Enter number of characters: 6
Enter characters:
a b c d e f
Enter frequencies:
5 9 12 13 16 20
```

```
Huffman Codes:
a : 000
b : 001
e : 01
f : 10
c : 110
d : 111
bharg@HP:~$
```